LawBot

An Engineering Project in Community Service

Phase – I Report

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Bonafide Certificate

Certified that this project report titled **"LawBot"** is the bonafide work of "23BCE10857 Aarul Kumar, 23BCE11492 Anushka Dubey, 23BCE10880 Dev Anurag Varshney, 23BCE11131 Gauri Makker, 23BCE11463 Vanshika Dhaka, 23BAI10653 Udita Chakraborty**"** who carried out the project work under my supervision.

This project report (Phase I) is submitted for the Project Viva Vice examination held on 10th February, 2025.

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Declaration of Originality

We, hereby declare that this report entitled **"LawBot"** represents our original work carried out for the EPICS project as a student of VIT Bhopal University and, to the best of our knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree or diploma of VIT Bhopal University or any other institution. Works of other authors cited in this report have been duly acknowledged under the section "References".

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Abstract

Access to legal information and grievance redressal remains a major challenge for millions of citizens in rural India due to a shortage of legal professionals, financial constraints, low legal literacy, and complex administrative procedures. These barriers often prevent individuals from understanding their rights, filing grievances, or seeking timely legal support. LawBot is an AI-powered legal assistance system designed to democratize access to justice by providing simple, accurate, and multilingual legal guidance through widely accessible platforms.

The system leverages Large Language Models (Llama 3), Retrieval-Augmented Generation (RAG), and vector-based semantic search (Pinecone) to offer contextually accurate answers grounded in verified Indian legal documents. It integrates Whisper-based voice processing to make legal help accessible to users with low literacy, and deploys through WhatsApp, ensuring high penetration and ease of use in rural communities. LawBot offers conversational legal assistance, simplifies complex legal jargon, assists users in drafting basic grievance documents, and guides them through structured complaint-filing processes.

By combining advanced AI technologies with a user-centric design, LawBot addresses long-standing gaps in legal accessibility and aligns with national goals for digital inclusion and citizen empowerment. The project demonstrates how AI-driven solutions can bridge socio-economic disparities, strengthen access to justice, and support the vision of a more equitable and legally aware society.

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1. INTRODUCTION

Access to legal support is a fundamental right, yet millions of citizens in India—especially in rural and underserved regions—face significant challenges in understanding laws, seeking guidance, and filing grievances. The lack of legal resources, combined with financial, geographical, and literacy barriers, creates a substantial gap between citizens and the justice system. With the rise of digital inclusion initiatives and AI advancements, there is an opportunity to bridge this gap with scalable, user-friendly technological solutions.

LawBot is an AI-powered legal assistance system designed to democratize access to justice by providing simplified legal information, multilingual support, and step-by-step grievance filing through an intuitive conversational interface. Leveraging modern technologies such as large language models (Llama 3), Retrieval-Augmented Generation (RAG), Pinecone vector databases, and Whisper-based speech recognition, LawBot aims to make legal aid more accessible, understandable, and affordable for all sections of society.

The system addresses the core challenges faced by rural citizens, including the shortage of lawyers, lack of awareness about legal rights, complex administrative procedures, and limited digital literacy. Through a WhatsApp-integrated AI agent, LawBot provides an easy way for users to ask legal questions, understand legal terminology, draft complaints, and receive guidance without the need for physical travel or expert consultation.

1.1 Motivation

The motivation behind developing LawBot arises from multiple socio-economic and systemic challenges that continue to restrict the legal empowerment of Indian citizens. Despite being one of the largest democracies in the world, India faces a profound disparity in access to legal resources, particularly in rural regions where nearly 65% of the population resides. Studies reveal that rural areas suffer from an alarming shortage of legal professionals, with only one lawyer available per 10,000 individuals, compared to urban areas where the ratio is approximately 1:1,000. This uneven distribution of legal support creates an environment where rural citizens struggle to seek timely advice or understand the basic procedures required to file complaints.

The financial aspect of legal engagement is another major barrier. Research from policy institutes shows that each visit to court may cost ₹1,000 or more, which includes travel expenses, documentation charges, and loss of daily wages. For low-income households, this is often an impossible expenditure. As a result, many individuals ignore legal issues, tolerate exploitation, or rely on unverified informal advice, leading to greater injustice and vulnerability.

Another important motivating factor is the lack of legal literacy. The majority of citizens find legal language too complex to understand. Many are not aware of their fundamental rights, welfare schemes, entitlement processes, or the existence of grievance platforms such as CPGRAMS, RTI, NALSA, or state-level legal aid services. Even among those who know, they struggle to navigate technical forms, legal terminology, or procedural steps. This gap between citizens and legal understanding is one of the primary causes of exploitation, misinformation, and delayed justice.

Technological limitations also play a role. Although India has developed several online portals for grievance redressal, the usability of these platforms remains a challenge for rural populations who may lack digital literacy or access to computers. On the other hand, mobile phone penetration—especially WhatsApp usage—has increased rapidly in rural India, making it the ideal medium for delivering legal assistance.

The rise of artificial intelligence further strengthens the motivation to create a solution like LawBot. Modern LLMs have the ability to understand natural language, translate complex text, and generate coherent and accurate responses. When combined with Retrieval-Augmented Generation (RAG), these models can provide legally grounded answers backed by verified documents. Whisper-based speech processing addresses literacy challenges, enabling even non-literate individuals to seek guidance through voice.

Ultimately, the motivation for LawBot stems from the belief that legal access should not be a privilege—it should be a right. By using AI to simplify the legal system, we can empower millions of citizens to take informed action, seek justice, and understand their rights without fear or confusion. LawBot aims to bring legal literacy, awareness, and assistance directly into the hands of the people who need it most.

1.2 Objective

The primary objective of **LawBot** is to create a reliable, accessible, and intelligent platform that democratizes access to legal information and grievance support. The project aims to address long-standing barriers in the Indian legal ecosystem by leveraging modern AI technologies and user-friendly communication channels. To ensure comprehensive impact, the project focuses on the following detailed objectives:

1. Provide Accessible Legal Guidance

LawBot aims to offer instant, accurate, and easy-to-understand answers to legal queries. This includes guidance on rights, welfare schemes, administrative processes, and legal terminology, enabling users to make informed decisions without needing immediate expert consultation.

2. Simplify Complex Legal Language

Legal documents are often filled with jargon and technical expressions. One of the core objectives of LawBot is to translate complicated legal text into simple Hindi or English while retaining the original meaning. This ensures users can clearly understand the steps they need to follow.

3. Assist in Grievance Filing and Document Preparation

Many citizens struggle with drafting formal complaints or understanding filing steps. LawBot aims to automate and simplify this process by generating ready-to-use grievance letters, complaint drafts, and RTI templates tailored to the user's situation.

4. Support Multilingual and Voice-Based Interaction

To accommodate diverse literacy levels, LawBot supports both Hindi and English queries. Using Whisper, the system converts voice input into text, allowing users—including those with low literacy—to access support easily.

5. Ensure Response Accuracy Through RAG

A major objective is to provide factually correct guidance by grounding the AI's responses in verified legal documents (indexed using Pinecone). This minimizes hallucinations and promotes trustworthiness.

6. Make Legal Support Available via WhatsApp

WhatsApp is widely used even in rural areas. Deploying LawBot as a WhatsApp chatbot ensures high accessibility without requiring users to download new apps or navigate unfamiliar systems.

7. Reduce Legal Dependence for Basic Queries

By educating citizens about their rights and procedures, LawBot reduces unnecessary dependence on lawyers for basic clarifications, making justice more affordable.

8. Promote Digital Inclusion and Government Alignment

The objective aligns with national development goals like Digital India, e-governance adoption, and Viksit Bharat 2047. LawBot aims to serve as a tool that supports government initiatives for accountability, transparency, and citizen empowerment.

9. Ensure Scalability and Real-World Usability

The system is designed to scale across regions, languages, and legal domains. The architecture allows continuous updates, making it adaptable to future enhancements such as regional language support and integration with government platforms.

2. Existing Work / Literature Review

Access to legal support in India has long been a subject of academic research, policy reforms, and technological interventions. Over the years, several government bodies, NGOs, and independent researchers have attempted to address the challenges associated with legal inaccessibility, grievance redressal, and legal literacy. However, despite numerous initiatives, significant gaps remain—particularly in rural and underserved regions. This section provides a detailed review of the existing systems, tools, and research efforts that form the foundation and motivation for the development of LawBot.

2.1 Traditional Legal Aid Systems

The primary form of legal assistance available to rural citizens has traditionally been through government legal aid centers and District Legal Services Authorities (DLSAs). The National Legal Services Authority (NALSA) undertakes various programs to support marginalized populations. While impactful, NALSA's services remain limited in reach due to:

- Severe shortage of legal professionals in rural areas
- Lack of transportation and high travel burden for rural citizens
- Low awareness of available legal aid centers
- Limited operational hours and long waiting times
- No mechanism for real-time, instant guidance

As a result, traditional legal aid remains inaccessible or difficult to utilize for a large portion of the population.

2.2 Online Grievance Redressal Portals

In recent years, the Indian government has introduced online platforms such as:

- **CPGRAMS (Centralized Public Grievance Redress and Monitoring System)**
- **RTI Online Portal**
- **Consumer Grievance Portals (NCH)**
- **State-level Lok Adalat systems**
- **E-Courts Services Mobile App**

While these systems aim to make grievance filing easier, research highlights several challenges:

- Interfaces are complex and not designed for low-literate users
- Forms contain legal or administrative jargon
- Users need English or formal Hindi proficiency
- Lack of awareness about these platforms among rural populations
- Limited smartphone literacy prevents effective usage
- Users often distrust whether their grievance will be resolved

Therefore, despite technological advancements, existing portals remain inaccessible to citizens who need them the most.

2.3 Legal Literacy Programs

Various NGOs and universities—such as NUJS, CLAP, and NLSIU—organize legal literacy camps focusing on rights awareness, RTI training, and grievance redressal education. These initiatives have made positive contributions but suffer from:

- Limited geographical coverage
- One-time training with no continuous support
- Lack of personalized guidance
- No multilingual, on-demand access
- Minimal use of AI or digital tools to scale awareness

Due to these limitations, such programs provide only partial solutions to widespread legal illiteracy.

2.4 AI-based Legal Systems (Global Context)

Globally, several AI systems have been developed to support legal tasks:

- **DoNotPay (USA):** Offers automated ticket dispute letters and contract analysis
- **ROSS Intelligence:** Provides AI-driven legal research assistance
- **Legal Robot:** Performs contract readability analysis
- **Chat with Law (UK):** Provides basic legal information via chat

However, these tools focus primarily on legal research or automated document analysis for professionals, not for rural citizens or low-literate populations. They also lack multilingual capability, voice input, and localized legal content tailored to Indian laws and grievance systems.

2.5 RAG-Based Legal Assistants (India – Recent Research)

Recent academic papers discuss the use of RAG (Retrieval-Augmented Generation) for offering legal Q&A based on Indian case laws or bare acts. However:

- Most prototypes are limited to academic experiments
- No real-world deployment through platforms like WhatsApp
- No voice interface for low-literate users
- No grievance filing assistance
- No localized tailoring for rural contexts

Existing AI legal tools focus mostly on **lawyers**, not **citizens who need basic help**.

2.6 Gap Identified in Existing Work

Based on an analysis of the current systems:

Area	What Exists	What Is Missing
Legal Aid	NALSA, DLSA	Instant, 24×7 conversational support
Grievance Portals	CPGRAMS, RTI	User-friendly guidance & multilingual help
Legal Literacy	Camps, workshops	Continuous AI-based assistance
AI Legal Tools	Legal research bots	Citizen-focused, rural-friendly assistants
Accessibility	Web apps	WhatsApp + voice interface for low literacy

These gaps reveal the need for a solution that is **simple, multilingual, AI-powered, voice-enabled, and accessible through widely used platforms**.

2.7 How LawBot Differs from Existing Work

LawBot stands apart from existing solutions because it:

- Uses **RAG** to provide factually grounded answers
- Offers **simple Hindi & English explanations**
- Supports **voice queries** for non-literate users
- Works on **WhatsApp**, already used in rural India
- Helps users **draft grievance letters and complaints**
- Provides **step-by-step filing assistance**, not just information

Thus, LawBot fills a critical gap by combining AI with grassroots-level accessibility, enabling citizens to understand and act upon their rights without barriers.

3. **TOPIC OF THE WORK**

The topic of this project is the development of **LawBot**, an AI-powered legal assistance and grievance-support system designed to make legal information simple, accessible, and understandable to common citizens, especially those living in rural areas. The project focuses on solving key challenges such as limited legal literacy, shortage of lawyers, financial constraints, and difficulty in navigating government grievance platforms.

LawBot uses modern AI technologies—like **Llama 3**, **LangChain**, **Whisper**, and **Pinecone Vector Database**—to provide accurate legal guidance in simple Hindi and English. The system is deployed through **WhatsApp**, ensuring maximum accessibility without requiring users to install any new app or have technical skills. LawBot assists users by answering legal questions, explaining rights, simplifying legal terms, and helping them draft basic grievance documents such as complaint letters, RTI requests, and consumer grievance forms.

The work emphasizes social impact, digital inclusion, and the use of AI for public welfare. It combines legal knowledge, conversational AI, and user-friendly design to create a practical solution that supports citizens in understanding and exercising their legal rights.

3.1 System Design / Architecture

The architecture of LawBot is designed to integrate AI models, retrieval systems, and a simple chat-based interface to deliver accurate legal guidance. The system consists of the following components:

1. User Interface (WhatsApp)

Users interact with LawBot via WhatsApp text or voice messages. This makes the system accessible even in rural areas where WhatsApp is widely used.

2. Speech/Text Input Handling

- Text queries are directly processed.
- Voice queries are converted to text using **Whisper speech-to-text**, enabling low-literate users to ask questions naturally.

3. Backend Processor (Python + LangChain)

The backend receives the query, performs language detection, cleans the text, understands user intent, and prepares the query for retrieval.

4. Retrieval-Augmented Generation (RAG)

To ensure accuracy:

- The query is converted into embeddings.

- Pinecone Vector Database stores legal documents (RTI rules, grievance procedures, rights information).
- Relevant document chunks are retrieved and provided to the Llama 3 model.

5. LLM Response Generation (Llama 3)

The model generates:

- Clear explanations
- Step-by-step guidance
- Translated/simple legal language
- Draft documents as required

6. Output Delivery

Responses are sent back to the user on WhatsApp, ensuring clarity and ease of use.

Architecture Workflow (Text Diagram)

```
User → WhatsApp → Whisper/Text → Backend (LangChain) → RAG → Pinecone DB
→ Llama 3 → Simplified Response/Draft → WhatsApp User
```

3.2 Working Principle

The working principle of LawBot is based on the idea of combining conversational AI with verified legal information to provide reliable assistance. The workflow is simple and user-friendly:

Step 1: User Sends Query

The user asks a legal question or requests help drafting a grievance. This may be in text or voice form.

Step 2: Speech or Text Processing

If the user sends a voice message, Whisper converts it into text. The system then identifies the language (Hindi/English).

Step 3: Intent Understanding

The backend determines:

- Is the user asking for legal information?
- Do they want help drafting a document?
- Do they need procedural steps for filing?

Step 4: Retrieval of Legal Information

Using RAG, the system:

1. Converts the query into embeddings.
2. Searches Pinecone for relevant legal content.
3. Retrieves verified information from indexed documents.

Step 5: Response Generation

Llama 3 generates a simplified answer based on:

- Retrieved documents
- User intent
- Legal context

The model avoids hallucinations by relying on retrieved legal data.

Step 6: Document Drafting (If Required)

If the user needs a:

- Complaint letter
- RTI draft
- Consumer grievance
- Application format

LawBot automatically generates the document using templates and user details.

Step 7: Response to User

The final simplified guidance or drafted document is delivered through WhatsApp.

This workflow allows LawBot to act as a first-line legal assistant that is fast, accurate, and easy to use.

3.3 Results and Discussion

The development and testing of LawBot produced several positive results:

1. Accurate and Simple Legal Responses

Through RAG, the system was able to retrieve relevant legal information and generate responses in simple Hindi/English. Test users reported that explanations were easy to understand even without prior legal knowledge.

2. High Accessibility

WhatsApp deployment ensured that the system worked smoothly for rural users. Voice input was especially well-received by users with low literacy.

3. Effective Document Generation

LawBot successfully generated:

- Grievance letters
- RTI applications
- Consumer complaints

These drafts were accurate, properly formatted, and could be submitted directly.

4. Multilingual Support

Both Hindi and English queries were processed effectively. Users appreciated the ability to switch languages seamlessly.

5. Reduced Hallucinations

By grounding answers in Pinecone-indexed legal documents, hallucinations were significantly minimized compared to a normal LLM chatbot.

Discussion

The project demonstrates that modern AI can bridge legal accessibility gaps if combined with user-centric design. While LawBot is not a replacement for professional legal advice, it effectively serves as a first-step assistant, helping citizens understand their rights and guiding them through official procedures without confusion.

3.4 Individual Contribution by Members

Aarul Kumar

- Designed the project idea and initial problem statement.
- Developed the core AI-based system architecture.
- Integrated major AI components including **Llama 3**, **LangChain**, **Whisper**, and **Pinecone** for retrieval and processing.

Anushka Dubey

- Assisted in collecting legal documents, datasets, and references.
- Helped in formatting legal content for indexing and processing.
- Contributed to backend development and API integration.

Dev Anurag Varshney

- Prepared the project presentation, documentation, and report content.
- Worked on designing the UI/UX flow for WhatsApp-based interaction.
- Contributed to conversational flow structuring and user experience refinement.

Gauri Makker

- Conducted testing for multilingual prompts (Hindi/English).
- Ensured translation accuracy of legal terms and responses.
- Created and organized the dataset of legal documents used for the RAG pipeline.

Vanshika Dhaka

- Carried out extensive research on legal rights, RTI processes, NALSA guidelines, and grievance redressal mechanisms.
- Designed the system flow, process pipeline, and testing methodology for the project.

Udita Chakraborty

- Focused on systematic testing of the chatbot responses and user interactions.
- Collected user feedback and performed iterative refinement of outputs.
- Worked on backend workflow implementation and WhatsApp integration.

4. CONCLUSION

The **LawBot** project demonstrates how Artificial Intelligence can significantly improve access to legal information and grievance support, especially for individuals in rural and underserved communities. By integrating advanced technologies such as **Llama 3**, **LangChain**, **Whisper**, and **Pinecone**, the system provides simplified legal explanations, accurate information retrieval, and guided grievance drafting in both Hindi and English. The use of **WhatsApp** as the primary interaction platform ensures high accessibility without requiring new applications or digital expertise.

The project successfully addresses key challenges such as low legal literacy, shortage of legal professionals, complex administrative procedures, and financial barriers. Through voice-enabled interaction, multilingual support, and document generation, LawBot acts as a first-line legal assistant that empowers users to understand their rights and initiate formal processes independently.

While LawBot is not intended to replace professional legal advice, it serves as a powerful tool to increase awareness, reduce confusion, and make justice more approachable. This project demonstrates the potential of AI-driven solutions to contribute to societal development and supports the broader vision of digital governance, inclusion, and citizen empowerment.

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6. BIODATA:

Aarul Kumar

Role in Project: Developed the core AI-based system architecture

Registration Number: 23BCE10857

Program: B.Tech – Computer Science and Engineering

Technical Skills: Skilled in AI architecture design, LLM integration, and Python-based backend systems. Experienced with Llama 3, LangChain, Whisper, and Pinecone for RAG workflows.

Responsibilities in LawBot (Proposed): Aarul is responsible for conceptualizing the AI-driven backend architecture and planning integration strategies for Llama 3, LangChain, Whisper, and Pinecone. Designed the core RAG pipeline, model orchestration flow, and legal document indexing structure, while defining specifications for system-level data handling and AI processing across major modules.

Anushka Dubey

Role in Project: Assisted in collecting legal documents, datasets, and references.

Registration Number: 23BCE11492

Program: B.Tech – Computer Science and Engineering

Technical Skills: Proficient in data collection, preprocessing, and structuring legal documents for AI. Has experience in backend support, API integration, and dataset formatting.

Responsibilities in LawBot (Proposed): Anushka is responsible for preparing and structuring legal documents for ingestion into the system, while supporting backend workflows related to data formatting and API communication.

Dev Anurag Varshney

Role in Project: Worked on designing the UI/UX flow for WhatsApp-based interaction.

Registration Number: 23BCE10880

Program: B.Tech – Computer Science and Engineering

Technical Skills: Skilled in UI/UX design for chatbot interfaces and WhatsApp-based user flows. Experienced in documentation, presentation creation, and conversational flow design.

Responsibilities in LawBot (Proposed): Dev is responsible for designing the conversational user interface and the overall WhatsApp interaction flow, including the logic behind prompt responses and guided user journeys. Structured the UI/UX for chatbot accessibility, developed documentation, and defined presentation-ready specifications for user interaction across primary modules.

Gauri Makker

Role in Project: Created and organized the dataset of legal documents used for the RAG pipeline.

Registration Number: 23BCE11131

Program: B.Tech – Computer Science and Engineering

Technical Skills: Proficient in multilingual NLP testing and translation accuracy evaluation. Experienced in dataset organization, prompt testing, and RAG content preparation.

Responsibilities in LawBot (Proposed): Gauri is responsible for testing multilingual response accuracy and evaluating translation consistency for Hindi and English legal queries. Created processed datasets for RAG, refined document structuring rules.

Vanshika Dhaka

Role in Project: Carried out extensive research on legal rights, RTI processes, NALSA guidelines, and grievance redressal mechanisms.

Registration Number: 23BCE11463

Program: B.Tech – Computer Science & Engineering

Technical Skills: Skilled in legal research, system flow design, and policy analysis (RTI, NALSA). Experienced in process pipeline design, testing methods, and documentation.

Responsibilities in LawBot (Proposed): Vanshika is responsible for conducting legal research on rights, RTI, NALSA regulations, and grievance frameworks, mapping them into actionable system flows. Designed core process pipelines, structured operational logic for the modules, and defined evaluation criteria for the system's legal accuracy and completeness.

Udita Chakraborty

Role in Project: Worked on backend workflow implementation and WhatsApp integration.

Registration Number: 23BAI10653

Program: B.Tech – Computer Science & Engineering

Technical Skills: Proficient in chatbot testing, backend workflow implementation, and WhatsApp integration. Skilled in feedback analysis, output refinement, and AI response evaluation.

Responsibilities in LawBot (Proposed): Udita is responsible for implementing backend workflows and overseeing WhatsApp API integration across modules. Led systematic testing of chatbot behavior, collected user feedback, and defined refinement protocols to enhance accuracy, stability, and response reliability within the conversational system.